Response to the Reviewers

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Embedded Artificial Intelligence in Battery Management Systems:
Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation

We sincerely thank the editor and all reviewers for their careful assessment and constructive comments. The following document provides a point-by-point response and identifies the corresponding changes in the revised manuscript.

1 Response to Reviewer #1

Manuscript: *Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation*

The page and line references below refer to the final 62-page line-numbered anonymous revision compiled on 22 July 2026. Figure captions and other float contents are identified by figure number and page because the `lineno` package does not assign reliable line numbers inside floats. No new STM32 measurements, battery experiments, HPC training runs, random-seed study, or validation on another chemistry were performed. Added analyses use the existing model artefacts, exported C implementations, test data, and recorded benchmark results. The revised manuscript states these boundaries explicitly. All references below were refreshed after integration of the Reviewer 2, Reviewer 3, and Reviewer 4 revisions.

1.1 General response

We thank the reviewer for the detailed comments. We revised the method descriptions, removed unsupported causal claims, documented the exact scope of the energy estimate and mixed-precision quantization, and added five supplementary analyses. These analyses cover pruning diagnostics, temporal stability, SOH filtering, utility-weight sensitivity, and transient software input-buffer faults. We have carefully distinguished observations from the evaluated models and replay from conclusions that would require new chemistry, training, power-measurement, or broad hardware fault-injection campaigns.

1.2 Comment 1: Cell-chemistry generalisation

> This study only evaluated the estimator on the LFP DoE cycle aging dataset containing a single battery chemical system. The authors need to discuss whether the observed trade-off between accuracy and compression can be generalized to other chemical systems with different degradation characteristics, such as NMC or LCO.

Response: We agree that the original manuscript did not delimit chemistry transferability clearly enough. The revised Discussion now separates architecture-level storage effects from model accuracy. For a fixed architecture, removing the same number of hidden channels or storing the same recurrent matrices as INT8 produces comparable parameter-count changes independent of chemistry. However, the accuracy-compression trade-off depends on chemistry, operating range, degradation trajectory, and training data. Our numerical MAE and preferred compression level are therefore demonstrated only for the evaluated JGC LFP cells at 25 degrees Celsius and cannot be transferred to NMC or LCO without retraining and validation. We added relevant NMC SOC/SOH literature to show feasibility of compact data-driven models while explicitly stating that those studies do not validate our numerical trade-off.

Changes in the manuscript: general discussion of transfer across cell chemistries (pages 39–40, lines 838–848); platform and scope discussion (page 40, lines 849–876); Conclusion and Outlook (pages 41–42, lines 877–917).

1.3 Comment 2: Quantization-aware and pruning-aware training

> The pruning and quantization experiments were conducted on the fully-precision floating-point trained model. The authors need to investigate whether quantization-aware training or pruning-aware training, compared to post-compression training, can further improve the trade-off between accuracy and efficiency.

Response: We clarified that the present study evaluates two deployable transformations of trained FP32 models: one-shot structured pruning followed by brief post-pruning fine-tuning, and post-training recurrent-weight quantization. Quantization-aware training, pruning-aware training, and an iterative pruning schedule were not performed because they require a new training campaign. The revised paper no longer suggests that the tested variants represent the best attainable jointly trained compression result. It explains why training-aware compression may recover accuracy and identifies this comparison as future work.

Changes in the manuscript: structured magnitude pruning and the explicit post-pruning fine-tuning boundary (pages 17–20, lines 387–440); post-training mixed-precision quantization details (pages 20–22, lines 441–480); Discussion of the compression boundary and training-aware comparison (page 40, lines 849–856); future work (page 42, lines 908–917).

1.4 Comment 3: Evidence for the claimed pruning regularisation effect

> The SOC estimator showed a slight improvement in MAE after pruning, which is attributed to the regularization effect. The authors need to provide evidence of weight distribution or unit activation statistics to support this regularization explanation.

Response: We agree that the original causal interpretation was too strong. We removed the claim that pruning acted as a regulariser. The revised text reports the lower SOC MAE as an observation for one trained model and test split. We added an L2 unit-saliency ranking and recurrent-weight distribution comparison. The 19 removed SOC units occupy the lowest saliency ranks by construction, which verifies implementation of the stated criterion. The figure is explicitly described as diagnostic evidence, not proof of improved generalisation. We also state that repeated seeds, cross-validation, and a matched fine-tuning ablation were not available.

Changes in the manuscript: pruning criterion and diagnostic scope (pages 18–19, lines 394–434); Error Distribution Analysis and removal of the causal regularisation claim (page 31, lines 643–660); Discussion (page 37, lines 761–779); Appendix A.1 (page 43, line 922) and Fig. A.16 (page 44).

1.5 Comment 4: Scope of the inference-energy values

> The inference energy consumption is based on hardware measurement reports. The authors need to clarify whether this includes the energy consumed by memory access or only the core computation. In microcontroller deployments, flash access may dominate energy consumption.

Response: We corrected the terminology throughout. On-device kernel time was measured with the STM32 DWT cycle counter. Energy was not measured independently for each model. The reported value is now defined as the proxy `E_est = 0.5 W * measured inference time`, where 0.5 W is an assumed representative average power for the complete development board. The timed kernel interval includes the instructions and memory accesses executed by the LSTM and MLP, but the constant-power calculation cannot separate CPU computation, flash access, SRAM access, UART, regulator losses, or other board consumers. It therefore cannot establish whether flash access dominates actual consumption. The Methods, KPI tables, utility score, Discussion, and Conclusion consistently call this value an estimate or timing-derived proxy. The revised Abstract focuses on measured flash and inference-time outcomes rather than presenting the proxy as independent energy evidence.

Changes in the manuscript: measured resource results in the Abstract (page 1, lines 17–23); measured kernel interval and build settings (page 25, lines 538–558); energy-proxy definition and interpretation (pages 25–27, lines 559–589); utility definition and sensitivity results (pages 34–35, lines 711–732); Discussion (pages 38–39, lines 814–837); Conclusion (page 41, lines 877–894). The revised KPI table captions and headings appear in Tables 1 and 2 on pages 26–27.

1.6 Comment 5: Temporal dependence and very long sequences

> The streaming replay test used the complete test set and did not consider the temporal dependence of predictions. The authors need to evaluate whether the compressed model maintains stability on very long sequences, as quantization errors may accumulate.

Response: We added a temporal analysis of the full-stream output trajectories. The naturally ordered replay is divided into ten consecutive equal-count segments without resetting recurrent states. We report target MAE and P95 error in each segment and the mean absolute deviation of each compressed output from Base. Quantized-to-Base SOC deviation changes from 0.332 percentage points in the first tenth to 0.271 percentage points in the final tenth, so the evaluated output does not exhibit quantization-specific monotonic drift. SOH deviations fluctuate with ageing phase and likewise show no monotonic quantization accumulation. We limit this conclusion to the recorded replay and do not claim stability for arbitrary sequence lengths.

Changes in the manuscript: host-side temporal-analysis method (page 23, lines 492–503); Temporal Stability and Filter Interaction results (pages 29–30, lines 618–642); conclusion (pages 41–42, lines 895–907); Appendix A.2 and Fig. A.17 (page 45).

1.7 Comment 6: SOH filter design, cutoff frequency, delay, and compression

> The SOH estimator operates on the filtered signal. The authors need to record the filter design and cutoff frequency and analyze how compression interacts with filtering to affect the overall estimation delay and accuracy.

Response: We documented the complete two-stage causal pipeline. Every raw model trajectory is first aligned to the initial SOH label. Stage 1 applies a symmetric step limiter and an EMA with `alpha_1 = 0.02`. Stage 2 consumes the stage-1 output, applies an EMA with `alpha_2 = 1e-6`, and limits downward changes to `2e-8` per sample. At 1 Hz, the EMA-only equivalent values are: stage 1 time constant 49.5 s, cutoff 3.22 mHz, and 90% response time 114 s; stage 2 time constant 11.57 days, cutoff `1.59e-7 Hz`, and 90% response time 26.65 days. The nonlinear limiters have no single cutoff frequency.

We also evaluated the exported Base, Pruned, and Quantized C models on a common ordered input sequence and report the raw, stage-1, and final sequential outputs and MAE. The results demonstrate that filtering changes both absolute error and model ranking, while the strong final stage introduces substantial delay. This controlled comparison is used specifically to examine the interaction between compression and filtering and does not replace the primary benchmark values.

Changes in the manuscript: SOH prediction post-processing subsection (pages 14–15, lines 307–326); SOH Results text (pages 27–28, lines 608–617); compression/filter comparison (pages 29–30, lines 618–642); Discussion (page 37, lines 773–779); conclusion (pages 41–42, lines 895–907); Appendix A.3 and Fig. A.18 (page 46). The SOH dashboard caption appears in Fig. 11 on page 30.

1.8 Comment 7: Utility weights and ranking sensitivity

> The comparison between pruning and quantization variants used the normalized utility score U. The authors need to clarify the weight factors used in this score and conduct a sensitivity analysis to show how the ranking changes with the variation of application priorities.

Response: We generalised the utility equation to four explicit non-negative weights for accuracy, flash, RAM, and estimated energy, constrained to sum to one. The originally reported score uses equal weights of 0.25. We evaluated all 1,771 combinations on a 0.05 grid and added focused sweeps in which one objective receives 25–85% weight and the remainder is divided equally. Pruned is top-ranked for 94.64% of SOC combinations and 53.36% of SOH combinations; Base wins 0% and 28.85%, and Quantized wins 5.36% and 17.79%, respectively. Thus, SOC Pruned is robust across most tested priorities, while the SOH ranking changes when accuracy or flash dominates. We also clarify that the energy objective is a timing-derived proxy and is not statistically independent of inference time under the fixed-power assumption.

Changes in the manuscript: revised utility equation, explicit equal weights, and sensitivity results (pages 34–35, lines 711–732); interpretation of the purpose and decision value of the score in the Discussion (page 39, lines 823–837); Appendix A.4 and Fig. A.19 (page 47).

1.9 Comment 8: Robustness to embedded-system faults

> The current limitations mentioned the lack of fault injection research. The authors need to assess the robustness of the compressed model to common embedded system faults, which may affect the correctness of inference.

Response: We added two deliberately limited software-level fault analyses. Randomly unavailable output reports are handled with hold-last while the underlying inference trajectory continues without interruption. More importantly, a stateful input-buffer experiment now evaluates whether a transient fault propagates through the recurrent estimator. The exported C implementations of Base, Pruned, and Quantized process a continuous 8000-sample segment from the evaluation trajectory of one representative LFP cell at 1 Hz. For each task and model, 30 independent events are distributed over the sequence after at least 2048 clean samples. Ten events affect voltage, ten current, and ten temperature. FP32 bits are indexed from zero at the least significant bit. Bit 22 is therefore the highest-order bit of the 23-bit fraction field. Flipping this bit produces a pronounced but finite input change without changing the sign or exponent. At each event, it is flipped in one input value for exactly one sample. The clean and disturbed branches start from the same recurrent state, after which both receive the same 60 unmodified inputs without resetting the LSTM state.

We report three complementary properties. The disturbed-clean peak quantifies the immediate fault effect, a sustained threshold determines recovery time, and the change in target MAE quantifies the resulting accuracy loss. The evaluation window contains the corrupted sample and 60 subsequent clean samples. It therefore contains 61 samples while covering a 60-s recovery horizon at 1 Hz. This separation is necessary because a transient can accidentally move an imperfect prediction closer to the target and because a window MAE alone can hide one short peak. All SOC variants recover in all 30 events within 60 s. Their median peak deviations are 3.29–3.82 percentage points, P95 peak deviations are 16.18–23.81 percentage points, and P95 increases in window MAE are 1.42–1.59 percentage points. For SOH, median peaks are 0.83–1.59 percentage points and P95 peaks are 3.37–5.85 percentage points. The shares recovered within 60 s are 93.3% for Base, 90.0% for Pruned, and 86.7% for Quantized. SOH P95 increases in window MAE remain 0.053–0.122 percentage points. No non-finite output occurs in the evaluated events.

This experiment directly tests short-term recurrent propagation after a transient software input-buffer upset, but it remains a bounded sensitivity analysis. It does not inject faults into physical sensors, communication links, weights, activations, recurrent-state memory, or MCU memory and does not support a hardware-safety claim. These broader fault classes remain future work.

Changes in the manuscript: Limited Software-Level Fault Sensitivity method and MAE definition (page 24, lines 511–537); Limited Software Fault Sensitivity results (pages 36–37, lines 735–760); interpretation in the Discussion (page 40, lines 865–876) and Conclusion (pages 41–42, lines 895–907); Appendix A.5 and Fig. A.20 (page 48), with the model-specific target-MAE values in Table A.4 (page 49).

2 Response to Reviewer #2

Manuscript: *Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation*

The page and line references below refer to the final 62-page line-numbered anonymous revision compiled on 22 July 2026. No new model training, hardware measurement, calculation, or figure was required specifically to address these comments. The revised literature positioning for Reviewer 2 used verified references already contained in the manuscript bibliography; two additional Zotero-verified references were subsequently added for Reviewer 4. All page and line references in the responses to Reviewers 1, 3, and 4 were refreshed after the final integration.

2.1 General response

We thank the reviewer for the constructive comments concerning the positioning and presentation of the work. We substantially rewrote the Introduction and Related Work sections, formulated one concise objective followed by three explicit contributions, and separated the demonstrated novelty from claims that are outside the scope of the study. We also rewrote the Conclusion and Outlook to quantify the principal results, state their practical meaning and limitations, and identify concrete future work. Finally, the manuscript received a language and logic pass, with particular attention to the Abstract, Introduction, Related Work, Results, Discussion, and Conclusion.

2.2 Comment 1

> The novelty of the work must be clearly addressed and discussed, compare your research with existing research findings and highlight novelty, (compare your work with existing research findings and highlight novelty).

Response: We agree. The revision now distinguishes the contribution from both battery-estimation studies and generic model-compression studies. The Related Work section compares the present benchmark with prior embedded SOC and SOH implementations, including studies reporting microcontroller timing and memory and recent work on compact or quantized estimators. It also contrasts the present battery-specific, continuous-state benchmark with a generic pruning-and-quantization benchmark. This comparison identifies the remaining gap: existing work generally reports either estimation accuracy, one embedded implementation, or one compression method, but does not provide the same paired, auditable comparison of structured pruning and recurrent-weight quantization for both SOC and SOH under one STM32 measurement protocol. We now state explicitly that the novelty lies in this end-to-end comparative deployment benchmark and its transparent evidence boundaries, not in proposing a new recurrent cell or a new compression algorithm.

Changes in the manuscript: motivation, research gap, objective, and contribution statement (pages 4–5, lines 78–126); expanded comparison with prior embedded and compression studies (pages 5–8, lines 127–216, especially pages 7–8, lines 168–216); contribution statement in the Conclusion and Outlook (pages 41–42, lines 877–917).

2.3 Comment 2

> The main objective of the work must be written on the more clear and more concise way at the end of introduction section.

Response: We agree. The end of the Introduction now contains one direct objective sentence: the study quantifies how structured hidden-unit pruning and recurrent-weight post-training quantization change the accuracy, memory footprint, and inference cost of stateful SOC and SOH estimators under identical data and deployment conditions on the same STM32 platform. This sentence is followed by three compact contributions covering the auditable model-to-firmware path, the controlled Base–Pruned–Quantized comparison, and the joint accuracy/resource analysis.

Changes in the manuscript: concise objective and three-item contribution statement at the end of the Introduction (page 5, lines 111–126).

2.4 Comment 3

> Introduction section must be written on more quality way, i.e. more up-to-date references addressed. Research gap should be delivered on more clear way with directed necessity for the conducted research work.

Response: We agree. The Introduction was restructured to move from the current SOC/SOH estimation context to embedded constraints and then to the specific need for a common compression benchmark. The literature discussion now includes recent 2024 and 2025 review and embedded-deployment studies already present and verified in the project bibliography, alongside the relevant foundational work. We also explain why cross-study accuracy values cannot be compared without considering chemistry, data split, target construction, and error definitions. The resulting research gap motivates the study directly: deployment decisions require accuracy, Flash, RAM, timing, and implementation precision to be reported under one controlled protocol. The verified bibliography was sufficient to address this comment. The two additional references requested by Reviewer 4 were subsequently added and are documented in that response.

Changes in the manuscript: rewritten Introduction with current context and embedded motivation (pages 4–5, lines 78–126); updated and more critical literature synthesis and explicit research gap (pages 5–8, lines 127–216).

2.5 Comment 4

> Conclusion section is missing some perspective related to the future research work, quantify main research findings.

Response: We agree. The Conclusion and Outlook was rewritten. It now reports the principal changes in Flash, RAM, inference time, and MAE separately for SOC and SOH and for both compression variants. It distinguishes measured inference time from the constant-power energy proxy and explains the practical operating points: structured pruning is the more balanced option for the evaluated firmware, whereas recurrent-weight quantization is mainly attractive when Flash is the dominant constraint and its mixed-precision runtime overhead is acceptable. The conclusion also states the limits of transfer to other chemistries, controllers, initialisations, and fault classes. Future work is prioritised around kernel optimisation and direct power profiling, broader validation across seeds, temperatures, chemistries, and controller families, training-aware compression, and internal embedded fault injection.

Changes in the manuscript: revised Conclusion and Outlook (pages 41–42, lines 877–917); quantified findings (page 41, lines 884–894); engineering interpretation and transferability (pages 41–42, lines 895–907); prioritised future work (page 42, lines 908–917).

2.6 Comment 5

> English language should be carefully checked and carefully check paper for language typos.

Response: We agree. We performed a language and logic pass throughout the manuscript. This included correcting grammar and terminology in the Abstract, rewriting long and repetitive passages in the Introduction and Related Work, clarifying the independence of the SOC and SOH estimators in the Methods, removing formulaic and unsupported wording from the Results, and rewriting the Discussion and Conclusion for more direct technical interpretation. We also standardised terms such as `mean absolute error`, `inference time`, `estimated energy`, `FP32`, and `INT8` according to their defined meanings.

Representative changes in the manuscript: revised Abstract (page 1, lines 6–23); rewritten Introduction and Related Work (pages 4–8, lines 78–216); clarified model-output wording (page 12, lines 261–269); revised Results opening and interpretation (pages 27–35, lines 590–732); revised Conclusion and Outlook (pages 41–42, lines 877–917).

3 Response to Reviewer #3

Manuscript: *Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation*

The page and line references below refer to the final 62-page line-numbered anonymous revision compiled on 22 July 2026. Figure captions and other float contents are identified by figure number and page because the `lineno` package does not assign reliable line numbers inside floats. No new STM32 measurements, model-training runs, random-seed study, cross-validation, or benchmark on another controller family were performed. The quantitative analyses used to address the present comments rely on the evaluated architectures, inspected C firmware, archived model artefacts, and previously recorded DWT kernel times. The manuscript now states these evidence boundaries explicitly. All references below were refreshed after the final integration of the revision.

3.1 General response

We thank the reviewer for identifying several places where the implementation and the scope of the conclusions required clearer documentation. We revised the abstract, feature-engineering description, pruning and quantization methods, STM32 benchmark protocol, Discussion, limitations, and Outlook. Four supplementary figures were added for analytical model-complexity scaling, the scope of the L2 pruning criterion, the verified mixed-precision boundary, and static quantized-runtime accounting. Where the requested experiment was not available, we removed unsupported causal or transferability claims and state precisely what can and cannot be concluded from the evaluated models and firmware.

3.2 Comment 1: Model complexity and hardware transferability

> The abstract reports that pruning reduces flash footprint and energy per inference by approximately 40%, yet it does not clarify whether these savings remain consistent across different model complexities (e.g., varying hidden dimensions) or across other STM32 families (e.g., F4 series). Given the diversity of embedded AI deployments, it would strengthen the paper to briefly discuss the model sensitivity and hardware transferability of these energy savings, or at least delineate the scope within which the current conclusions hold, thereby improving the generalizability of the findings for practitioners working on different BMS platforms.

Response: We agree and have separated architecture-level scaling from hardware-specific measurements. For input dimension `D`, hidden size `H`, and MLP width `M`, the revised manuscript gives the analytical count

`N_MAC(H) = 4H(D + H) + HM + M.`

For a general hidden-unit pruning fraction `p`, the retained size is `H_p = (1-p)H`. The resulting analytical reduction is

`R_MAC(H,p) = [4(2p-p^2)H^2 + p(4D+M)H] / [4H^2 + (4D+M)H + M],`

and its large-model limit is `2p-p^2`. The revised Figure A.21 therefore no longer presents 30% as an isolated calculation. It shows the complete relation between pruning fraction and asymptotic operation reduction. For example, pruning fractions of 10%, 20%, 30%, 40%, and 50% correspond to large-`H` limits of 19%, 36%, 51%, 64%, and 75%, respectively.

The evaluated `p=0.30` point gives 22,080 to 12,124 MACs for the implemented SOC change from 64 to 45 hidden units and 85,120 to 46,208 MACs for the SOH change from 128 to 90 hidden units. These finite-model reductions are 45.1% and 45.7%, respectively, because the linear and fixed terms remain relevant at the implemented dimensions. The 30% pruning fraction was used as a single moderate operating point that produced a substantial dimensional reduction while retaining useful estimator accuracy. It was not selected by an exhaustive rate sweep and is not claimed to be optimal. The manuscript now identifies a systematic comparison of separately pruned and fine-tuned rates as a pruning-focused follow-up rather than implying that the present operating point is universally preferable.

The revised Abstract now links the reported savings explicitly to the evaluated 30% pruning configuration and reports measured flash and inference-time reductions rather than presenting the timing-derived energy estimate as independent evidence. These measured ratios apply to the evaluated models, model dimensions, STM32H753ZI firmware, and GCC `-O0` build. Clock rate, FPU/DSP support, compiler optimisation, cache, and memory organisation can change the flash and timing ratios. We therefore make no numerical transfer claim for STM32F4 or another controller family without a new hardware benchmark.

Changes in the manuscript: bounded Abstract statement (page 1, lines 13–23); general analytical MAC model and implemented hidden-size reductions (pages 15–16, lines 327–362); rationale and scope of the evaluated 30% pruning fraction (page 19, lines 411–434); inspected firmware and build settings (page 25, lines 538–558); measured/static comparison and implementation interpretation (page 38, lines 784–813); general hardware-transfer discussion (page 40, lines 857–864); Conclusion and Outlook (pages 41–42, lines 877–917); Appendix A.6 (page 50) and Fig. A.21 (page 51).

3.3 Comment 2: Derivative implementation and non-ideal sampling

> The paper employs centered finite differences for voltage and current derivatives but does not address efficient implementation on embedded platforms, nor does it discuss numerical stability under variable sampling rates or data loss. In real-world BMS applications, sensor noise and communication delays are inevitable. It would be valuable to elaborate on the robustness strategies employed when computing derivatives on the STM32, or to suggest alternative filtering approaches that could maintain numerical stability under non-ideal conditions, thereby ensuring reliable real-time operation.

Response: An audit of the preprocessing code showed that the original term "centered finite differences" was incorrect. The evaluated features use timestamp-aware backward differences,

`Delta t_k = max(t_k - t_(k-1), epsilon_t)`,

`dU/dt = (U_k - U_(k-1)) / Delta t_k`, and analogously for current.

This formulation is causal, stores only the preceding sample and timestamp, and has constant work per update. However, the derivative and cumulative channels in the reported STM32 benchmark were prepared on the host and transmitted as part of the complete six-feature UART vector. The benchmark deliberately isolates the LSTM/MLP inference kernel; its DWT times do not include acquisition or feature construction. We now document this boundary instead of implying that centered differences were calculated on the microcontroller.

Because variable sampling, missing measurements, derivative noise, and causal derivative filtering or limiting were not evaluated as embedded modules in this compression study, we do not claim measured robustness for those conditions. The limitations now state that a production feature front end would require a separate validation of timestamp checks, missing-sample handling, and suitable causal filtering or limiting.

Changes in the manuscript: corrected timestamp-aware backward-difference equations, causal implementation cost, and host/STM32 boundary (pages 10–12, lines 245–269); DWT measurement interval and scaler placement (page 25, lines 538–558); scope and benefit of separating feature construction from inference (page 40, lines 865–876). No derivative-process figure was included because this implementation boundary is expressed more precisely by equations and text than by a process diagram.

3.4 Comment 3: Rationale for the L2 pruning score

> The pruning score is defined using L2 norms, but the authors do not justify why this criterion was chosen over alternatives such as gradient-based sensitivity or activation distribution, nor do they provide experimental comparisons. Given that pruning strategies directly impact the accuracy-efficiency trade-off, it would enhance the paper to include comparative experiments or at least provide a theoretical rationale for the chosen method, thereby helping readers assess its suitability for LSTM-based battery estimators and facilitating informed design decisions in similar embedded applications.

Response: We expanded the rationale for the gate-group L2 saliency score. It combines the four gate rows belonging to one hidden unit, is data independent after training, and maps directly to removal of complete dense rows, recurrent columns, and the associated MLP input. It therefore yields a smaller dense LSTM kernel rather than an irregular sparse mask. Gradient-based sensitivity would require training data and a backward pass, while activation-based ranking would require a representative calibration stream and an aggregation rule across time.

We did not run an experimental comparison of pruning criteria. The revised manuscript states this explicitly and does not claim universal superiority of L2 ranking. Figure A.22 provides a method-property rationale and the observed cross-task L2 rankings, while the existing diagnostics in Fig. A.16 show that the intended low-saliency channels were removed from the evaluated model.

Changes in the manuscript: expanded criterion definition, implementation rationale, and comparison scope (pages 17–20, lines 387–434); model-specific diagnostic interpretation (page 31, lines 643–660); Discussion (page 37, lines 761–779); Appendix A.7 and Fig. A.22 (page 52). The supporting model diagnostics remain in Appendix A.1 (page 43, line 922) and Fig. A.16 (page 44).

3.5 Comment 4: Weight-only mixed-precision quantization

> The paper applies per-row symmetric quantization exclusively to weights, while biases and activations remain in floating-point precision. Although this simplifies implementation, it may forgo additional opportunities for reducing memory and computational overhead. It would be insightful to discuss whether activation quantization was evaluated, or to explain the specific impact of retaining floating-point activations on both accuracy and hardware efficiency. Such clarification would help readers understand the design trade-offs and assess whether alternative quantization schemes could yield better resource-accuracy balances for BMS deployments.

Response: We verified the deployed SOC and SOH source paths against the exported headers and now describe the implementation consistently as weight-only mixed precision. Only the recurrent `W_ih` and `W_hh` matrices are stored as INT8 with FP32 row scales. Biases, hidden and cell states, gate activations, the MLP weights and computation, and the output remain FP32. The recurrent matrices constitute approximately 80% of the raw Base model constants, which motivated targeting them first.

Retaining FP32 activations and states avoids introducing another quantization source into the recurrent update, but it preserves FP32 runtime-state storage and arithmetic. The evaluated implementation therefore cannot provide the RAM and latency benefits of a full-integer path. Activation quantization was not evaluated, and the accuracy contribution of retaining FP32 activations was not isolated experimentally. We identify activation/state quantization and an optimised integer kernel as future work rather than implying that the current implementation exhausts the available efficiency gains.

Changes in the manuscript: exact precision boundary and its consequences (pages 20–22, lines 441–480); resource interpretation (pages 33–34, lines 672–710); quantized-kernel interpretation and possible optimisations (page 38, lines 789–813); future-work scope (page 42, lines 908–917); Appendix A.8 and Fig. A.23 (page 53).

3.6 Comment 5: Interpretation of the lower pruned SOC MAE

> The SOC error distribution reveals that the pruned model achieves slightly better MAE than the baseline, which the authors attribute to a regularization effect. However, this observation may be sensitive to training data splits or model initialization. It would be prudent to investigate whether this advantage persists across different random seeds or cross-validation settings, or to provide a theoretical explanation for how pruning could enhance generalization. Such analysis would prevent overinterpretation of what might be a dataset-specific phenomenon and strengthen the claims about pruning's benefits.

Response: We agree that the original causal interpretation was not supported by the available evidence. The manuscript no longer attributes the lower SOC Pruned MAE to a regularisation effect. It now reports a 0.35-percentage-point improvement only as an observation for the evaluated model and test split. The L2 saliency and recurrent-weight diagnostics verify which channels were selected, but they cannot demonstrate that pruning caused improved generalisation.

No repeated-seed, cross-validation, or matched unpruned fine-tuning experiment was available, so we do not claim persistence across model initialisations or data partitions. This limitation is stated in the Results, Discussion, limitations, Conclusion, and future-work paragraph.

Changes in the manuscript: non-causal Results interpretation (page 31, lines 643–660); model- and split-specific Discussion (page 37, lines 761–779); broader-validation discussion and conclusion (pages 39–42, lines 838–907); future random-seed validation (page 42, lines 908–917); supporting diagnostic Fig. A.16 (page 44) and criterion-scope Fig. A.22 (page 52).

3.7 Comment 6: Root causes of quantized inference time

> The increased inference time of quantized models is noted but not deeply analyzed regarding its root causes, such as dequantization overhead, changes in memory access patterns, or cycle consumption from floating-point conversions. Providing performance profiling data-including cycle counts for key operations and memory bandwidth utilization-would help identify specific bottlenecks. Additionally, discussing potential optimizations like operator fusion or hardware acceleration would offer practical guidance for future work aiming to improve quantized LSTM deployment on resource-constrained microcontrollers.

Response: We expanded the implementation-level analysis while keeping static accounting separate from measured profiling. Base and Quantized have the same topology and therefore the same model MAC counts: 22,080 for SOC and 85,120 for SOH. In the inspected quantized C expressions, each recurrent INT8 weight is converted to FP32 and multiplied by its row scale inside the innermost accumulation. This adds 17,920 source-level FP32 scale multiplications for

SOC and 68,608 for SOH. Counting one model MAC and one additional scale multiplication as one operation gives a static Quantized/Base ratio of about 1.81 for both tasks.

The measured DWT kernel-time ratios are 4.99 for SOC and 1.29 for SOH. Their disagreement with the identical static ratio shows why a simple operation count cannot explain cross-kernel timing. The count omits cast cost, loop and index overhead, loads/stores, nonlinear functions, cache/flash behaviour, and compiler effects. The evaluated firmware was built at `-O0` and does not use an optimised integer dot-product path. Reduced recurrent-weight traffic may offset more arithmetic overhead in the larger SOH model, but memory bandwidth was not measured, so this remains an implementation-based interpretation rather than a causal attribution.

We did not have operation-level DWT traces or memory-bandwidth measurements and therefore do not present the new analysis as cycle-level hardware profiling. The revised Discussion instead identifies concrete future optimisations: move shared row scales outside the innermost accumulation where mathematically valid, fuse scale and bias operations, use DSP-optimised or integer dot-product kernels, and quantize activations and recurrent states for a complete integer path. Figure A.24 places the static source-level counts beside the previously measured total kernel times and labels the evidence types explicitly.

Changes in the manuscript: measurement interval, scaler placement, compiler and FPU settings (page 25, lines 538–558); mixed-precision KPI interpretation (pages 25–27, lines 559–589); latency interpretation (pages 33–34, lines 672–710); static counts, measured ratios, omitted costs, and optimisation options (page 38, lines 789–813); future-work paragraph (page 42, lines 908–917); Appendix A.9 and Fig. A.24 (page 54).

4 Response to Reviewer #4

Manuscript: *Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation*

The page and line references below refer to the final 62-page line-numbered anonymous revision compiled on 22 July 2026. Figure captions and other float contents are identified by figure number and page because the `lineno` package does not assign reliable line numbers inside floats. No new battery experiment, model-training campaign, or STM32 measurement was performed for these comments. The two papers explicitly suggested by the reviewer were imported from the author's Zotero bibliography and are cited as Refs. [16] and [17]. Figures 2 and 7 retain the established visual language of the original manuscript with clarified captions, while Figure 6 provides the revised pruning schematic.

4.1 General response

We thank the reviewer for the detailed comments on presentation, literature context, and mathematical clarity. We shortened and refocused the Abstract, added and analysed the two suggested references, reorganised the method description around the actual Base, Pruned, and Quantized transformations, and added implementation-level equations for pruning, quantization, mixed-precision inference, and evaluation metrics. We also divided long paragraphs, performed a manuscript-wide language and logic pass, clarified the captions of the general design-of-experiments and quantization schematics, and revised the pruning schematic.

4.2 Comment 1

> The expression of the abstract should be improved. The length should be suitable for the abstract, not too long.

Response: We agree. The Abstract was rewritten close to the original narrative rather than as a list of qualifications. It introduces the engineering application and implemented artificial intelligence, establishes the full-precision estimators as the accuracy reference, and then presents the long-horizon and hardware-in-the-loop results. The reported flash and inference-time savings are tied explicitly to the evaluated 30% pruning configuration, while the accompanying analytical model relates the computational reduction to pruning fraction and network dimensions. Detailed qualification of the timing-derived energy estimate and transferability is retained in the Methods and Discussion, where it can be explained without interrupting the Abstract. All acronyms are defined at first use. The title contains no acronym, and the keyword list has been reduced to six written-out terms.

Changes in the manuscript: revised Abstract and keywords (page 1, lines 6–26; Abstract lines 6–23 and keywords lines 24–26).

4.3 Comment 2

> Please conduct more reference analysis for the research topic, such as An improved parameter identification and radial basis correction-differential support vector machine strategies for state-of-charge estimation of urban-transportation-electric-vehicle lithium-ion batteries, Improved singular filtering-Gaussian process regression-long short-term memory model for whole-life-cycle remaining capacity estimation of lithium-ion batteries adaptive to fast aging and multi-current variations, and so on.

Response: We added both suggested articles from the author's Zotero export and analysed their relationship to the present work. The revised Related Work explains that the first combines online parameter identification, adaptive radial-basis correction, and a differential support-vector machine for SOC estimation under dynamic vehicle tests, while the second combines singular filtering, Gaussian-process regression, and an LSTM for whole-life capacity estimation under fast ageing and varying currents. We then state the relevant distinction: these studies broaden estimation methods and operating conditions, whereas neither reports the paired post-training pruning/quantization comparison with linked-firmware memory and recurrent microcontroller timing investigated here. They are therefore treated as complementary evidence, not as direct benchmark substitutes.

Changes in the manuscript: new comparative literature analysis and citations [16,17] (page 6, lines 143–154); full bibliography entries with DOI metadata (pages 57–58, lines 1022–1035).

4.4 Comment 3

> The expression logic can be improved for your proposed method so that the innovation can be clearly understood. More mathematical analysis should be conducted to clarify the methods.

Response: We reorganised the methods to distinguish three operations explicitly: the common FP32 Base equations, structured pruning that changes the hidden dimension and dense parameter submatrices, and quantization that changes the stored representation of selected recurrent weights without changing topology. The revised text now gives the analytical MAC expression and derives the general relative reduction for a pruning fraction `p`, including its large-model limit `2p-p^2`. It then gives the retained-index submatrix construction in Eq. (17), the exact symmetric per-row quantizer in Eqs. (18)–(20), the reconstruction-error bound in Eq. (21), and the mixed-precision STM32 gate equation in Eq. (22). It also states which quantities remain FP32 and why storage reduction does not imply fully integer execution. These additions clarify that the contribution is an auditable paired deployment benchmark rather than a new recurrent cell or compression algorithm.

Changes in the manuscript: common architecture and general analytical MAC scaling (pages 15–16, lines 327–362); structured pruning definition and dense submatrix construction (pages 17–20, lines 387–434; Fig. 6 on page 20); exact quantization and mixed-precision equations (pages 20–22, lines 441–480; Fig. 7 on page 22); explicit benchmark objective and contributions (pages 4–5, lines 78–126).

4.5 Comment 4

> Grammar should be checked and improved for the entire content. Please try to make every sentence correct and easy to understand.

Response: We performed a manuscript-wide language pass and simplified ambiguous or overly compressed sentences. Particular attention was given to the Abstract, Introduction, Related Work, preprocessing definitions, pruning and quantization methods, Results interpretation, Discussion, Limitations, Conclusion, and declarations. We also corrected an author-name encoding error and removed duplicated acknowledgement text.

Representative changes in the manuscript: Abstract (page 1, lines 6–23); Introduction and Related Work (pages 4–8, lines 78–216); Methods (pages 9–27, lines 220–589); Results and Discussion (pages 27–40, lines 590–876); Conclusion and Outlook (pages 41–42, lines 877–917).

4.6 Comment 5

> Please pay more attention to the expression logic and ensure the logic is correct for every sentence.

Response: We agree and revised the argument flow from motivation to evidence boundary. The Introduction now closes with one objective and three contributions; Related Work ends with the precise research gap; the methods separate data, targets, common architecture, compression transformations, and measurement definitions; the Results distinguish observations from causal claims; and the Discussion separates measured findings, implementation-based interpretations, and unevaluated transferability. Unsupported statements, including a causal regularisation explanation for the lower Pruned SOC MAE and independent energy-measurement wording, were removed or bounded.

Changes in the manuscript: objective and contribution sequence (page 5, lines 111–126); research gap and implementation boundary (pages 7–8, lines 168–216); Base/Pruned/Quantized method boundary (pages 15–22, lines 327–480); evidence-based interpretation and transferability discussion (pages 37–40, lines 761–876); Conclusion and Outlook (pages 41–42, lines 877–917).

4.7 Comment 6

> Some paragraphs are too long. Please optimize your expression.

Response: We split and tightened long paragraphs throughout the manuscript. The revised structure separates dataset design from check-up procedures, training settings from deployment state handling, benchmark replay from firmware measurement, static operation counts from timing interpretation, and demonstrated limitations from future work. This reduces sentence load while preserving the quantitative details required for reproducibility.

Representative changes in the manuscript: dataset and preprocessing (pages 9–15, lines 220–326); training and deployment descriptions (pages 15–17, lines 327–386); benchmark methodology (pages 23–27, lines 481–589); Discussion and transferability (pages 37–40, lines 761–876); Conclusion and Outlook (pages 41–42, lines 877–917).

4.8 Comment 7

> Some figures are unclear. Please try to optimize them.

Response: We revised the presentation of three explanatory figures. Figure 2 uses the general three-factor design-of-experiments geometry from the original manuscript, while its caption now explains the generic condition labels and directs the reader to the dataset-specific factor levels in the text. Figure 6 depicts the dense hidden-channel pruning path and the implemented SOC and SOH dimension changes. Figure 7 restores the more detailed FP32-to-INT8 deployment schematic from the original manuscript. Its caption clarifies that the INT8 data type spans `-128...127`, while the implemented symmetric quantizer generates `-127...127` and leaves `-128` unused. It also states that only the recurrent matrices are quantized and that the remaining operations stay in FP32.

Changes in the manuscript: restored general Fig. 2 with clarified caption (page 10); revised Fig. 6 and caption (page 20); restored detailed Fig. 7 with implementation-consistent caption (page 22).

4.9 Comment 8

> More expression about the mathematical analysis should be conducted to show your idea clearly. Please try to highlight your proposed method and focus on it.

Response: In addition to the method equations described in our response to Comment 3, we formalised the evaluation criteria so the relation between predictions and reported percentages is unambiguous. The revised manuscript defines signed percentage-point error, MAE, RMSE, and the empirical P95 threshold in Eq. (24) and the following expression. The pruning analysis now derives the finite-model reduction as a function of both hidden size and pruning fraction and proves the large-model limit `2p-p^2`. Figure A.21 visualises this relation from 0% to 60% pruning and highlights the evaluated 30% point, whose asymptotic limit is 51%. It also distinguishes these architecture-level counts from measured flash, latency, and the timing-derived energy proxy. This mathematical treatment focuses the contribution on a transparent, reproducible comparison of two deployment transformations.

Changes in the manuscript: analytical MAC count and general pruning-fraction reduction (pages 15–16, lines 335–351); pruning equations and operating-point scope (pages 18–19, lines 394–434); quantization and reconstruction equations (pages 21–22, lines 448–472); formal error metrics (pages 25–26, lines 559–564); measured/static interpretation (page 38, lines 784–813); analytical scaling in Appendix A.6 (page 50) and Fig. A.21 (page 51).